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Optical Music Recognition for Full Music Sheets: A CRNN-CTC Approach for Monophonic Lines

GEP Deliverable 1: Context and Scope

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1 Introduction and Context

1.1 Academic/Professional Framework

This project is developed in the academic context of the Bachelor’s Thesis (TFG) at FIB–UPC, under modality A, as part of the requirements to complete the Computer Science degree at the Universitat Politècnica de Catalunya (UPC). The project is supervised by Manel Frigola Bourlon, professor at the Facultat d’Informàtica de Barcelona (FIB), and guided by Xavier Morales Sorolla as GEP tutor.

The work is framed within the Computació specialty. The topic was selected due to my interest in the intersection of music and technology, and because of the potential impact of Optical Music Recognition (OMR) in areas such as music education, digital archiving, and music information retrieval.

1.2 Domain Concepts and Terminology

Optical Music Recognition (OMR): The process of automatically converting scanned or photographed music sheets into machine-readable symbolic formats (e.g., MusicXML, MIDI).

Music Sheet/Score: A printed or digital document containing musical notation, including notes, staves, time signatures, and other musical symbols.

End-to-End System: A unified approach that maps input directly to output without intermediate manual steps, typically using a single neural network or an integrated pipeline.

Music Information Retrieval (MIR): The interdisciplinary field focused on extracting meaningful information from music data.

Symbolic Notation: A machine-readable representation of music (e.g., MusicXML, MIDI), in contrast to audio waveforms.

Digital Archiving: The process of converting and preserving documents in digital form for long-term accessibility.

In addition to these core concepts, the project relies on basic music theory (notation, rhythm, and structure) and technical foundations in deep learning, computer vision, and data processing. When required, these notions will be introduced and explained to keep the document accessible to readers with different backgrounds.

1.3 Problem Statement

As a musician, I have repeatedly faced the need to digitize paper scores for sharing, archival purposes, and use in composition or analysis software. This is especially relevant in rehearsal and live-performance contexts, where musicians may receive a physical chart and need to perform it immediately.

In these situations, paper limits flexibility. Performers may need to transpose parts for different instruments, generate quick audio playback to understand style and groove, or store the score in a searchable digital library (e.g., on a tablet). Manual transcription is slow and error-prone, which motivates the need for an efficient and accurate method to convert physical music sheets into machine-readable formats.

1.4 Stakeholders

- **Musicians:** Readers and performers who benefit from digital scores for practice, performance, sharing, and transposition.

- **Music Educators:** Teachers who use digital sheet music in class and provide interactive learning resources to students.
- **Music Archivists:** Professionals responsible for preserving and organizing collections, who benefit from improved accessibility and long-term preservation.
- **Music Software Developers:** Companies and independent developers who can integrate OMR outputs into notation, playback, or learning applications.
- **MIR Researchers:** Researchers in music information retrieval who can use OMR data for analysis, recommendation systems, and musicology studies.
- **General Public:** Music enthusiasts interested in accessing and interacting with digital music sheets for personal use and learning.

2 Justification of the Chosen Alternative

2.1 State of the Art / Existing Alternatives

Current OMR literature distinguishes three broad families of approaches:

1. **Traditional modular pipelines (e.g., rule-based systems):** They decompose the task into fixed stages (pre-processing, staff detection/removal, symbol segmentation, symbol classification, and reconstruction). Their main weakness is error propagation between stages, and they require significant manual engineering for each new score style or notation condition.
2. **CRNN-CTC approaches:** They combine convolutional feature extraction with recurrent sequence modeling (typically a bidirectional LSTM) trained with Connectionist Temporal Classification (CTC) loss, adapting methods originally developed for handwriting and text recognition. They are well-suited for monophonic staff lines where a left-to-right sequential assumption holds naturally, and they are lighter and faster to train than attention-based counterparts.
3. **End-to-end transformer approaches:** They formulate OMR as image-to-sequence translation with attention-based encoder-decoder architectures. This family handles long-range dependencies and complex page layouts well, making it the most promising direction for polyphonic and full-page scores, at the cost of higher data and compute requirements.

2.2 Comparison Criteria

The alternatives are compared using criteria aligned with the scope of this TFG:

- **Accuracy and robustness:** Resistance to image noise, quality variation, and notation complexity.
- **Polyphony support:** Ability to represent simultaneous notes and multiple voices correctly.
- **Layout awareness:** Capacity to process complete pages with multiple systems and non-trivial reading order.
- **Semantic completeness:** Generation of outputs that preserve musical meaning (pitch, duration, meter, and key context).

- **Implementation feasibility:** Compatibility with available time, compute resources, and development effort for a Bachelor’s thesis.
- **Evaluation compatibility:** Possibility of benchmarking with public datasets and reproducible metrics.

2.3 Comparative Analysis

Traditional pipelines provide interpretability and low entry cost, but their stage-by-stage structure tends to amplify upstream errors and requires substantial manual tuning for new score conditions. They remain useful as baselines and for understanding the problem structure.

CRNN-CTC models are a natural fit for the monophonic setting. A CNN encodes local visual features from the staff image, a bidirectional LSTM captures sequential context, and CTC decoding aligns the output sequence without needing explicit segmentation boundaries. This is exactly the problem structure of a single melody line such as those found in lead sheets (e.g., the Real Book), where reading order is strictly left-to-right and polyphony is absent.

Transformer-based models are the most powerful option for full-page and polyphonic transcription, but they demand substantially more annotated data and compute. They represent a natural extension once the monophonic problem is solved, rather than the primary implementation target for this TFG.

2.4 Selected Alternative and Rationale

The selected approach is a **CRNN-CTC model** for **monophonic music line recognition**, with the primary evaluation target being lead-sheet scores from **The Real Book (C edition)**.

Rationale:

1. **Well-matched to the monophonic setting:** Real Book melodies consist of single-voice staff lines with a strict left-to-right reading order, which is exactly the regime where CTC-based sequence models excel.
2. **Feasible within TFG constraints:** A CRNN-CTC is significantly lighter to train than a full transformer, requiring less annotated data and less GPU memory, which is important given the limited compute budget of a Bachelor’s thesis.
3. **Strong prior art:** CRNN-CTC architectures have been validated for OMR in systems such as the Calvo-Zaragoza monophonic OMR line recognizer and related work, providing reproducible baselines.
4. **Clear evaluation path:** Recognition at the staff-line level can be directly benchmarked with Symbol Error Rate (SER) and sequence-level metrics on standard or custom-annotated Real Book data.
5. **Natural extension to polyphony:** Once monophonic recognition is solid, the architecture can be progressively extended, or complemented with a transformer-based model, to address polyphonic staves and full-page structure.

Traditional modular pipelines and transformer-based models will be studied and discussed to provide context and comparison, but the main implementation is the CRNN-CTC approach described above.

2.5 Constraints and Decision Boundaries

- **Data availability:** Ground truth for Real Book pages is not publicly available in annotated form; the project will require building or adapting a dataset, potentially combining synthetic generation with manual annotation of a subset of pages.

- **Monophonic-first scope:** The initial system will target single-voice melodic lines only. Polyphony (chords, multiple voices) and full-page layout parsing are considered extensions, not primary deliverables.
- **Computational resources:** Training must fit available GPU memory and time. A CRNN-CTC is better suited to this constraint than large transformer models.
- **Sequence representation:** A compact symbolic encoding (e.g., a simplified ****kern**-like or ABC-like token sequence or even a custom one) will be used for the output, balancing expressiveness and training efficiency.
- **Project scope:** The objective is a solid and reproducible prototype that correctly transcribes monophonic lead-sheet lines, not a production-ready universal OMR engine.
- **Polyphony and full-page extension:** These are considered future work if time and resources allow, potentially employing a transformer-based model in a second phase.

3 Project Scope

3.1 General Objective

To develop a proof-of-concept CRNN-CTC model capable of transcribing monophonic music lines into a symbolic format, with the Real Book as the primary target corpus, and to explore potential extensions toward polyphonic recognition, demonstrating the feasibility of this approach within the constraints of a Bachelor's thesis.

3.2 Specific Objectives

- Survey and document the state of the art in OMR, covering traditional modular pipelines, CRNN-CTC models, and transformer-based end-to-end approaches.
- Implement a traditional modular pipeline as a baseline, using classical computer vision techniques for staff detection and symbol segmentation.
- Build or adapt an annotated dataset of monophonic staff lines derived from Real Book pages, including synthetic augmentation to increase sample diversity and quantity.
- Design, implement, and train a CRNN-CTC model for monophonic staff-line recognition, defining a compact symbolic output encoding suitable for melody representation.
- Evaluate the CRNN-CTC model against the baseline using standard OMR metrics (e.g., Symbol Error Rate) and analyse failure modes.
- Investigate and prototype, if feasible within scope and time, an extension of the system to handle limited polyphony or chord symbols commonly found in lead sheets.

3.3 Functional Requirements

- The system shall accept as input a scanned or photographed image of a music sheet page and detect the individual staff lines present on it.
- For each detected staff line, the system shall produce a symbolic sequence representing the musical content (pitches, durations, rests, accidentals, time and key signatures) in a compact, machine-readable format.
- The output format shall be sufficient to reconstruct the melody of a monophonic line unambiguously (e.g., reconstruct MIDI or import into notation software).

- The system shall process Real Book lead-sheet pages as the primary evaluation scenario.
- The system shall include a traditional modular pipeline baseline against which the CRNN-CTC model can be compared quantitatively.
- The system shall provide evaluation outputs (metric scores and error analysis) for the trained model on a held-out test set.

3.4 Non-Functional Requirements

- **Accuracy:** The CRNN-CTC model should achieve a Symbol Error Rate (SER) clearly below that of the modular baseline on the test set, demonstrating a meaningful improvement.
- **Reproducibility:** All experiments must be reproducible from a public repository with documented dependencies, dataset preparation scripts, and training configuration files.
- **Computational feasibility:** Training must complete in a reasonable time on a single consumer-grade GPU (e.g., NVIDIA RTX-class). Inference on a single staff line must run in well under one second.
- **Modularity:** The codebase should be structured so that the staff detection, line cropping, and recognition stages are independently testable and replaceable.
- **Portability:** The implementation should run on standard Linux environments using open-source libraries (PyTorch, OpenCV, etc.) without proprietary dependencies.
- **Documentation:** Code and experimental results must be documented clearly enough to allow the project supervisor and evaluators to understand and verify the work.

3.5 Scope Boundaries and Exclusions

In scope:

- Monophonic melody lines like the ones found on the Real Book pages.
- Staff-line detection and individual line cropping as a preprocessing stage.
- Design, training, and evaluation of a CRNN-CTC model for single-voice line recognition.
- Implementation of a traditional modular pipeline as a comparative baseline.
- Dataset construction, including synthetic data generation and/or manual annotation of a Real Book subset.
- Quantitative evaluation using standard OMR metrics (SER, pitch accuracy, rhythm accuracy).
- A survey and discussion of transformer-based OMR approaches as context and potential future direction.

Out of scope (for this TFG stage):

- Full polyphonic transcription (multiple simultaneous voices or complex chords beyond simple chord symbols).
- Full-page layout analysis (reading order recovery across multiple systems, lyrics, annotations).

- Audio synthesis or playback integration beyond basic MIDI export.
- Real-time or mobile deployment.
- Support for handwritten scores or non-Western notation systems.
- Training or fine-tuning large-scale transformer models from scratch.

3.6 Risks and Potential Obstacles

- **Limited ground-truth data:** Annotated Real Book data is not freely available. Building a sufficient training set manually is time-consuming. *Mitigation:* Use synthetic score generation (e.g., via LilyPond or MuseScore) to bootstrap training data, and apply data augmentation (noise, blur, rotation) to improve generalisation.
- **Staff detection errors propagating downstream:** If staff lines are incorrectly detected or cropped, the recogniser receives corrupted input. *Mitigation:* Dedicate early project phases to validating the detection stage independently before coupling it to the recognition model.
- **Encoding design complexity:** Defining a compact yet expressive symbolic encoding for the CTC output vocabulary may require iteration. *Mitigation:* Start with a minimal vocabulary (pitch class + octave + duration) and extend incrementally.
- **Model underfitting or CTC misalignment:** CTC training can be sensitive to vocabulary size and sequence length ratios. *Mitigation:* Follow established CRNN-CTC OMR recipes from the literature and monitor CTC loss curves carefully during training.
- **Scope creep toward polyphony:** Attempting to address polyphony prematurely could divert time from solidifying the monophonic core. *Mitigation:* Treat polyphony as a clearly gated extension, only pursued after the monophonic system meets the defined accuracy targets.
- **Computational resource constraints:** Limited GPU access could slow training iteration cycles. *Mitigation:* Use lightweight model configurations, mixed-precision training, and cloud compute credits if available.

4 Methodology and Rigour

4.1 Project Methodology

Describe the chosen methodology (e.g., classical/cascade, agile, hybrid) and justify why it fits the project.

4.2 Adaptation to This Project

Explain how the methodology is tailored to your concrete TFG context and constraints.

4.3 Execution Approach

Describe the intended phases/iterations at a high level and the expected deliverables/results of each one.

4.4 Monitoring and Follow-up Tools

Describe how progress will be monitored (meetings, milestones, indicators, planning board, repositories and logs).

5 Document Organisation

Briefly explain the structure of this document and what each section contains.

6 Conclusions

Summarize the context, chosen alternative, defined scope, and methodological approach. Highlight the expected contribution of the TFG.